WORKED SOLUTION

STEP III 1997, Question 11

Mechanics • Rigid Body Rotation

STEP III 1997, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q11

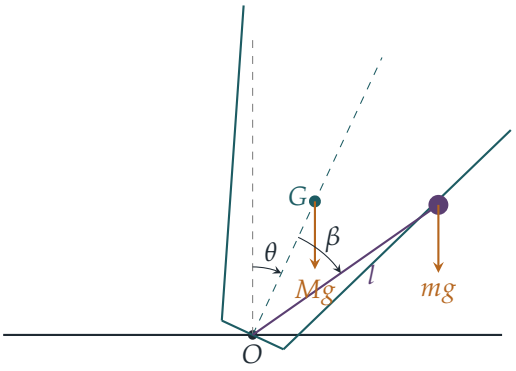
The key idea

While they touch, the bell and clapper turn about the same point O with the same angular velocity, so the only thing that can pull them apart is a difference in angular *acceleration*. Ask how fast each would accelerate on its own: the clapper and bell separate at the instant the bell, swinging freely, would accelerate just as fast as the free clapper. The contact force is then exactly zero, and the angular velocity never enters.

Solution

The model. The bell is a rigid body free to turn about the horizontal axis through O, with moment of inertia Mk^2 about it. The clapper is a particle of mass m on a light rigid rod of length l, also pivoted at O. The pivot is smooth and there is no air resistance. Measure every angle from the *upward* vertical, positive in the direction the clapper lies from the axis. Let θ be the angle of the bell's axis OG. While the two are in contact, the clapper rod stays at angle β to the axis, so it is at angle $\theta + \beta$.

Inside the bell, the clapper ball leans against the inner wall on the side towards which it would fall. So the wall pushes the ball back (towards smaller angles) and, by Newton's third law, the ball pushes the bell forwards. Let $\Gamma \geq 0$ be the moment about O of this contact force (including any friction). It acts as $-\Gamma$ on the clapper and $+\Gamma$ on the bell.



Equations of motion in contact. Taking moments about O for each body separately (the reaction at the pivot and the tension in the rod have no moment about O):

$$\text{clapper:}\quad ml^2\ddot{\theta} = mgl\sin(\theta+\beta) - \Gamma,$$
$$\text{bell:}\quad Mk^2\ddot{\theta} = Mgh\sin\theta + \Gamma.$$

The weight mg acts at distance l from O along a line at angle $\theta+\beta$ to the vertical, so its moment is $mgl\sin(\theta+\beta)$; similarly for Mg at G. Both bodies share the same $\ddot{\theta}$ because they move together.

Adding the equations eliminates Γ:

$$\ddot{\theta} = \frac{g(ml\sin(\theta+\beta) + Mh\sin\theta)}{ml^2 + Mk^2}.$$

Substituting back into the clapper equation and simplifying,

$$\Gamma = \frac{mMgl}{ml^2+Mk^2}\Big(k^2\sin(\theta+\beta) - hl\sin\theta\Big).$$

At release, $\theta = 0$ and $\Gamma = \dfrac{mMglk^2\sin\beta}{ml^2+Mk^2} > 0$, so the clapper does press on the bell at first, as it should.

> **Ask yourself** Why does separation happen exactly when Γ reaches zero, whatever the angular velocity?

The contact can only push, so it needs $\Gamma \geq 0$. While $\Gamma > 0$ the free clapper would accelerate faster than the bell, so it is held against the wall and the two move together. Because both turn about the same point O with the same angular velocity, the gap between the ball and the wall can only open through a difference in angular acceleration. Once Γ would turn negative, the bell's own acceleration $gh\sin\theta/k^2$ exceeds the clapper's $g\sin(\theta+\beta)/l$, and the wall swings away from the ball. So separation happens at the first θ with $\Gamma = 0$. (Such a θ always exists before $\theta = \pi$, since the bracket is $k^2\sin\beta > 0$ at $\theta = 0$ and $-k^2\sin\beta < 0$ at $\theta = \pi$.)

The separation angle. Put $\alpha = \theta + \beta$, the angle of the clapper rod with the upward vertical at separation. Then $\Gamma = 0$ gives

$$k^2\sin\alpha = hl\sin(\alpha-\beta) = hl(\sin\alpha\cos\beta - \cos\alpha\sin\beta).$$

Here $0 < \alpha < \pi$, so $\sin\alpha \neq 0$, and $\sin\beta \neq 0$. Dividing by $hl\sin\alpha\sin\beta$:

$$\frac{k^2}{hl\sin\beta} = \cot\beta - \cot\alpha,$$

that is,

$$\cot\alpha = \cot\beta - \frac{k^2}{hl}\operatorname{cosec}\beta,$$

as required. Notice that neither mass appears.

Checks

- *A short clapper.* As $l \to 0$ the correction term blows up, so $\cot\alpha \to -\infty$ and $\alpha \to \pi$. Sensible: a short pendulum has a large free angular acceleration $g\sin\alpha/l$, so a clapper on a very short rod keeps pushing the bell until $\sin\alpha$ is nearly zero, that is, until the pair is nearly upside down.
- *Equivalent form.* The condition is just $\dfrac{g\sin\alpha}{l} = \dfrac{gh\sin(\alpha-\beta)}{k^2}$: the free clapper and the free bell have equal angular accelerations.
- *Numerically.* With $k = 1$, $h = 0.6$, $l = 0.8$ and $\beta = 30°$, the formula gives $\cot\alpha = \sqrt{3} - 25/6 \approx -2.43$, so $\alpha \approx 157.7°$. Integrating the combined motion step by step from release, and stopping when Γ first hits zero, gives the same angle to many decimal places, for $M/m = 5$ and for $M/m = 500$ alike.

Common slips

- Using energy conservation to find $\dot\theta$ and then trying a "normal reaction" from circular motion. The angular velocity is common to both bodies and plays no part in whether they separate.
- Measuring α from the bell's axis rather than from the upward vertical. The rod is at β to the axis throughout the contact, so $\alpha = \theta + \beta$.
- Forgetting that Mk^2 is the moment of inertia about O, not about G. No parallel-axis correction is needed here.

Beyond STEP

This is a neat example of the general principle for finding when a constraint breaks: solve the constrained motion, compute the constraint force, and find where it changes sign. Lagrangian mechanics makes this systematic using Lagrange multipliers, which turn out to be the constraint forces. The same idea gives the classic particle leaving the surface of a sphere. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers rigid-body rotation about a fixed axis and is excellent preparation.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.